

EVOLUTION OF SUBJECTS OF TECHNICAL AND TECHNOLOGICAL DEVELOPMENT IN THE CONTEXT OF GLOBALIZATION OF THE WORLD ECONOMY

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ANNOTATION

The article analyzes the organizational structures evolution caused by the need to restructure a classical firm under the influence of technological progress. The main subjects of technical and technological development are identified and their functional role in the technical and technological renewal of production is shown. Based on the study of theoretical and methodological approaches of the classical school of political economy, Marxist direction, institutional theory, as well as evolutionary economics, the variability of ideas about the direct subject of technological changes is demonstrated. It is shown that in the course of evolutionary development, the main conductor of scientific and technological progress is a modern corporation, which combines research, investment and innovation resources into a single whole, turning into a powerful cluster of new technologies.

Keywords: inventions, innovations, investments, scientific and technological progress, technical and technological development, innovative firm, innovative entrepreneur, transnational corporation.

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Technical progress and its impact on the course of economic processes have always been in the focus of researchers' attention. In this regard, it is enough to pay attention to the fact that even in the stream of ideas of the classical school, there was established the view of technical progress as the result of improvements made by various subjects due to the deepening of the division of labor. Thus, A. Smith notes that "the invention of all those machines by which labour is so much facilitated and abridged seems to have been originally owing to the division of labour... A great part of the machines made use of in those manufactures in which labour is most subdivided, were originally the inventions of common workmen, who, being each of them employed in some very simple operation, naturally turned their thoughts towards finding out easier and readier methods of performing it... Many improvements have been made by the ingenuity of the makers of the machines, when to make them became the business of a peculiar trade; and some by that of those who are called philoso-

phers or men of speculation, whose trade it is not to do anything, but to observe everything; and who, upon that account, are often capable of combining together the powers of the most distant and dissimilar objects." [7, p.14]. That is, A. Smith brings to the forefront three subjects of technical progress - the worker, who directly rationalizes production due to direct contact with the means of production, the machine manufacturer and the scientist who invents these machines.

In K. Marx, the development of technology is considered to be more of a social phenomenon. First of all, the thinker says that the use of machines contributes to the growth of added value and, accordingly, profit, and also generates a "reserve army of labor" and lays the material basis for periodic crises of overproduction. However, in the scientist's reflections on the socio-economic role of technology, there are statements that also concern the subjects of technical progress. K. Marx draws attention to the figure of the inventor of machines, who creates a new model of technology, as well as the machine-building industry, which is born from the division of labor between manufactories and due to the growth in the number of that category of workers who, due to the semi-artistic nature of their occupations, become capable of producing new technology. "The inventions of Vaucanson, Arkwright, Watt and others could only be materialized because these inventors found a significant number of skilled mechanical workers," K. Marx points out [8, p. 393]. However, as history shows, these inventions did not create, according to K. Marx, the industrial revolution. "Just as an individual machine remains dwarfish as long as it is put into operation only by man, so a system of machines could not develop freely until the steam engine replaced the driving forces - animals, wind and even water - so the entire development of large-scale industry was paralyzed as long as the machine itself - the characteristic means of production of large-scale industry - owed its existence to personal strength, personal creativity, that is, it was dependent on the muscular strength, the accuracy of the eye and the virtuosity of the hands with which the individual worker inside the factory or the artisan outside it operates his dwarf tool" [8, p. 394].

The machine, from which the industrial revolution

originated, replaces the worker with a mechanism that simultaneously operates a set of identical or homogeneous means of labor, set in motion by a single motive force, whatever its nature. "In place of a separate machine comes this mechanical monster," concludes K. Marx, "whose body occupies entire factory buildings and whose demonic power, initially hidden in the almost solemnly measured movements of its gigantic components, breaks out in the feverishly frenzied dance of its numerous working organs" [8, p. 393]. So, to put into operation a system of machines based on steam energy, a new organizational unit of an industrial enterprise became necessary – and it became a factory – a large enterprise, which, thanks to the implementation of inventions and the improvement of technological processes in the production process, was able to significantly reduce production costs and achieve greater profitability due to this higher competitiveness. Intra-industry competition, described by K. Marx, served as an incentive for technical and technological renewal in individual factories and, ultimately, moved forward technical progress. At the same time, technical progress was viewed by K. Marx with a certain duality: as a progressive phenomenon in view of the growth of the well-being of society and, at the same time, as one that brings losses associated with the displacement of workers by machines and the growth of unemployment, harsh exploitation of the working class, periodic crises of overproduction. In neoclassical theory, the social component of technical progress is excluded from the scope of research, and the main attention is focused on the efficiency of using production factors within the firm. Neoclassical economists of the 19th century were interested, first of all, in the effect of reducing production costs and the level of prices that determined the profitability of business. Therefore, they did not pay attention to the technical and technological development itself, knowledge and inventions as its sources, but investigated how it can be accelerated through the activities of a separate firm. The founder of the neoclassical theory, A. Marshall, says that progress can be accelerated through thinking and work. Therefore, the improvement of machines and an increasingly deep division of labor contribute to the reduction of the cost of the process of manufacturing goods and, therefore, an important driving force of progress is the organization of production.

A. Marshall emphasizes that many of the types of savings from the use of specialized skills and machines are not always related to the size of factories. This especially applies to those types of savings that depend on the growth of the total amount of knowledge and technology on the scale of human civilization. "And though," writes the scientist, "he [small manufacturer] must al-

ways remain at a great disadvantage in getting information and in making experiments, yet in this matter the general course of progress is on his side: ... newspapers, and trade and technical publications of all kinds are perpetually scouting for him and bringing him much of the knowledge he wants ... Again, it is to his interest also that the secrecy of business is on the whole diminishing, and that the most important improvements in method seldom remain secret for long after they have passed from the experimental stage. It is to his advantage that changes in manufacture depend less on mere rules of thumb and more on broad developments of scientific principle; and that many of these are made by students in the pursuit of knowledge for its own sake, and are promptly published in the general interest. Although therefore the small manufacturer can seldom be in the front of the race of progress, he need not be far from it, if he has the time and the ability for availing himself of the modern facilities for obtaining knowledge." [9, pp. 79–80].

Thus, technical progress in neoclassical theory overcomes the limits of large-scale production and begins to be associated not so much with the scale of the production structure as with the resources of individual enterprises, their organization and management efficiency included in the number of factors of internal economy that contribute to technical progress.

"Side by side with the growing economy of skill, the growth of the business tends to bring with it economies of machinery and of all kinds of appliances; each successive improvement in machinery opens the way to further improvements, and each step in advance is likely to increase the advantage which the business has over its smaller rivals, and to diminish the price at which it can afford to sell its goods. This process may go on as long as the energies and enterprise of the man in control remain fresh, and as long as the risks of trade do not bring great losses..." [9, p.91]. Thus, the entrepreneur (businessman) becomes one of the main driving forces of technological progress, since he has the talent to effectively combine labor and capital in order to maximize profits.

This idea was further developed and even acquired new meaning in Joseph Schumpeter's concept of economic development. First and foremost, the scholar transformed the traditional understanding of production, emphasizing that it serves as a conduit for the new.

"To produce is to combine existing things and forces. To produce something different or in a different way is to create other combinations of these things and forces" [10, p.132] These other new combinations were later called innovations and it was with them that the driving forces of technical and technological development were associated. For the first time, the conduit of innovations

is not a separate enterprise, but a special subject - an entrepreneur-innovator. Joseph Schumpeter emphasizes that the implementation of innovations "is a special function and privilege of people, who are much fewer than those who would have such an opportunity in principle" [10, p. 149]. Such people, according to the researcher, are motivated to innovate by completely different motivations. While their activities are characterized by selfish motives and rationality, this type of entrepreneurs is not characterized by signs of hedonistic behavior. First, it is the desire for power, which comes from the need for self-realization. Second, it is the will to win – the desire to fight and achieve success for the sake of success itself. At the same time, the amount of profit here is considered not as a means of increasing individual wealth, but only as an indicator of success and a symbol of victory. Economic activity under such conditions turns into a kind of financial pursuit and is considered as a separate sport. And, finally, third, it is the joy of creativity, which becomes a defining moment of behavior. This includes both the simple satisfaction with the work done and the impetus for change for the sake of the change itself, for the sake of the opportunity to show courage and even for the sake of the difficulties that will have to be overcome [10, p.166]. So, the key factor in technical and technological development is the direct characteristics of the personality of the entrepreneur-innovator, who breaks the usual course of development and goes beyond the already known. It is the entrepreneur who begins to change the combination of production factors, improves them, directs them to a new combination, which should provide both better conditions for his activity and the realization of his internal motives. In turn, the introduction of the new undermines the usual course of the reproduction process, makes it impossible to use old production methods, devalues obsolete equipment, thereby causing the "creative destruction" of old methods of production and ensuring forward movement, technical progress of the economy.

However, in his work "Capitalism, Socialism, Democracy", examining the development of the capitalist economy in dynamics, Joseph Schumpeter considers this problem somewhat differently. The process of creative destruction is no longer associated exclusively with the activities of the innovative entrepreneur, but is made dependent on the organizational and legal type of enterprises. "The opening of new markets, internal and external, as well as the development of the organization from the craft workshop and factory to concerns... illustrates the process of economic mutation, which continuously revolutionizes the economic structure from the inside, disrupting the old and generating a new one. Such a process of creative destruction is the very

essence of capitalism...", writes the scientist [10, p.461].

Revealing the main driving forces of this process, Joseph Schumpeter draws attention to the fact that traditional competition based on price methods is being replaced by a new type based on the discovery of a new product, a new technology, a new source of raw materials and a new type of organization – a large-scale enterprise, which quite often occupies a monopoly position in the market. Paradoxically, such an organization of production in the process of creative destruction plays a progressive role. Indeed, when it comes not to a stationary state of the economy, but to its development under the influence of the emergence of new technologies and new goods, long-term investment is associated with a large share of uncertainty. Therefore, as forms of protection against an unpredictable future, firms choose patenting, preservation of technical and technological secrets, which for some time serve as a precautionary measure and allow them to build a long-term business strategy. Due to this, business entities are able to survive in conditions of a cardinal break in old structures and create new ones in the future. The facts show that the enterprise seeks to maximize the current net value of total assets, which is equal to the discounted value of the net income expected from them. This means, Schumpeter points out, that the enterprise will always introduce new methods of production if it is assumed that it will bring a greater flow of future income per unit of future investment [10, p.475].

This thesis is developed even more deeply within the institutional direction of economic theory. Its prominent representative J. K. Galbraith points out that for each given level of development and application of technology there is an optimal size of the firm – the size at which specialists, their knowledge, skills and the corresponding amount of capital are most rationally combined that should be sufficient for the development of innovations, as well as for the purchase of the necessary equipment. A small firm does not have such capital. "Even more significant," says J. K. Galbraith, "is that [the individual entrepreneur] does not possess the specialized scientific and technical personnel that are an inherent part of the modern organization and almost always indispensable for technical development. None of the most important technical achievements of recent times—atomic energy, the jet aircraft, the computer, or advances in electronics and agriculture—are the product of the individual entrepreneur. Ideas may still come from individuals. But, with few exceptions, only the organization can bring them to fruition." [11, p. 77]. Such an organization in the modern world is a large corporation, since it is this organizational structure that is able to concentrate the necessary capital and, importantly, direct it to innovation by maintaining a monopoly position.

In the conditions of modern development, which is characterized by globalization trends, the role of the leader of scientific and technological progress is transferred to powerful TNCs, which have both resource and institutional advantages to produce technological innovations. When studying international monopolization in the 1950s and 1960s, the 500 largest corporations were singled out; in the 1970s and 1980s – 200 of them, and today we are talking about the concentration of global economic power in the hands of 100 and even 50 largest monopolies. As a result, TNCs now control more than two-thirds of the main flows of scientific and technological knowledge (patents and licenses for new equipment, technologies, know-how), which is a sign of their growing role in interstate scientific and technological exchange. The key role of TNCs in monopolizing high-tech industries of global production has significantly increased in the conditions of the Industry 5.0 and becomes dominant in the conditions of the emergence of the technological order – Industry 6.0, when spending on innovative developments doubles every 3–4 years [12, p.75]. At the same time, national states actively contribute to strengthening of their TNCs, creating favorable conditions for them for technological leadership. The combination of incentives for the growth of the size of TNCs and the expansion of their sphere of influence creates an uneven situation in the world, generating technological gaps between different countries.

Thus, technical and technological development is based on three interrelated components - inventions, innovations and investments. Their combination is carried out by economic entities within the framework of organizational structures, which in different historical periods were represented by different firms. While at the beginning of the industrial revolution, inventors and direct investors in new equipment and technologies were separated in time and space, then in subsequent periods, a tendency to their combination was clearly distinguished. The modern corporation combines research, investment and innovation resources into a single whole, turning into a powerful cluster of

new technologies. The transnational nature of such corporations generates uneven technical and technological development in the world and becomes the basis for its relative stability.

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